

Generation Interconnection System Impact Study Report

For Queue Project Q13-0231

Magnesium Plant 230 kV

July 29, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0231. The Interconnection Customer proposes a 132.4 MW solar photovoltaic facility interconnecting at the Magnesium Plant 230 kV substation in the Illinois area. The request was received on April 22, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 171 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$3,966,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: NAPOLEON PEAKING STATION-RICHLAND 230.0 kV line
Contingency: TROY ENERGY LLC-RICHLAND 500.0 kV
Loading: 114.8% (Rating: 532 MVA)
DFAX: 13.8%
Pre-project loading: 111.4%
2. Facility: MASON ST 500-230 kV Transformer
Contingency: BEAVER-TREGO 500.0 kV
Loading: 105.2% (Rating: 1,081 MVA)
DFAX: 12.9%
Pre-project loading: 103.6%
3. Facility: CALDWELL-PINEY CREEK PROJECT 345.0 kV
Contingency: PINEY CRK PROJECT-HILLTOP 500.0 kV Line
Loading: 104.7% (Rating: 1,345 MVA)
DFAX: 9.9%
Pre-project loading: 103.8%
4. Facility: HIGHGATE-TROY ENERGY LLC 230.0 kV L
Contingency: MASON ST 500/230 kV TX
Loading: 104.5% (Rating: 534 MVA)
DFAX: 8.4%
Pre-project loading: 102.4%
5. Facility: TALENENERGY MARTINS CREEK LLC HARRISBURG 500-345 kV Auto
Contingency: CORONET-WESTWOOD GENERATION LLC 500.0 kV L
Loading: 101.2% (Rating: 1,183 MVA)
DFAX: 6.7%
Pre-project loading: 100.4%
6. Facility: GALION GENERATING STA 500 kV/230 kV TX
Contingency: TREGO-BEAVER SUBSTATION 500.0 kV Line
Loading: 100.5% (Rating: 916 MVA)

DFAX: 11.7%
Pre-project loading: 98.8%

7. Facility: ADIRONDACK SUBSTATION-JAMES DE YOUNG 230.0 kV
Contingency: ST. LAWRENCE-JAMES DE YOUNG 230.0 kV line
Loading: 100.4% (Rating: 546 MVA)
DFAX: 7.9%
Pre-project loading: 98.4%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
NAPOLEON PEAKING STATION-RICHLAND 230.0 kV line	Rebuild line with higher-capacity conductor	\$21,938,000	2% (\$543,000)
MASON ST 500 kV/230 kV XFMR	Replace transformer with higher-rated unit	\$20,427,000	2% (\$510,000)
CALDWELL-PINEY CRK PROJECT 345.0 kV	Reconductor line	\$19,098,000	3% (\$494,000)
HIGHGATE-TROY ENERGY LLC 230.0 kV L	Reconductor line	\$18,471,000	2% (\$461,000)
TALENENERGY MARTINS CREEK LLC HARRISBURG 500-345 kV Auto	Replace transformer with higher-rated unit	\$36,317,000	3% (\$951,000)
GALION GENERATING STA 500 kV/230 kV TX	Replace transformer with higher-rated unit	\$28,331,000	2% (\$707,000)
ADIRONDACK SUBSTATION-JAMES DE YOUNG 230.0 kV	Reconductor line	\$12,152,000	2% (\$300,000)
	Total allocated to project		\$3,966,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.